

NanoGT Match Report: Spinal Muscular Atrophy

Disease: Spinal Muscular Atrophy (ORPHA:70)

Primary gene: SMN1

Gene CDS: 891 bp

Inheritance: Autosomal recessive

Target tissues scored: CNS, muscle

Gene selection note: this disease has multiple listed genes (SMN1, SMN2); this report scores SMN1 only.

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype; Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Cell-autonomous protein across multiple tissues; high transduction coverage may be required.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: SMN protein deficiency from biallelic SMN1 mutation

Gene-addition compatibility: compatible

Preferred modality class: gene addition

Evidence level/status: direct / source_linked_needs_review

Evidence summary: FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene

Evidence source: FDA Zolgensma product page

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
- Endpoint risk: muscle/cardiac diseases may need functional, respiratory, imaging, or cardiac endpoints that progress slowly and vary by age/stage.
- Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	8.3/10	High	approved

Rank	Program	Vector	Score	Confidence	Approval
2	Zolgensma	AAV9	8.3/10	High	approved
3	AT132	AAV8	8.1/10	High	phase3
4	SRP-9001	AAV9	7.7/10	High	approved
5	BMN 307	AAV5	7.4/10	Medium	phase2

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 8.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)

- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: motor_neuron
- Disease mechanism: loss of function — SMN protein deficiency from biallelic SMN1 mutation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene
- Mechanism source: FDA Zolgensma product page (<https://www.fda.gov/vaccines-blood-biologics/zolgensma>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — SMN1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Therapeutic window is not wide; natural-history timing and irreversible damage should be reviewed
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 8.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	1.50	2.0	Same se- creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: motor_neuron
- Disease mechanism: loss of function — SMN protein deficiency from biallelic SMN1 mutation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene
- Mechanism source: FDA Zolgensma product page (<https://www.fda.gov/vaccines-blood-biologics/zolgensma>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected

- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — SMN1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Therapeutic window is not wide; natural-history timing and irreversible damage should be reviewed
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: AT132

Precedent disease: X-linked myotubular myopathy

Vector: AAV8

Tissue target: muscle

Composite score: 8.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist

Dimension	Score	Max	What it measures
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.10	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: muscle
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: myopathy
- Disease mechanism: loss of function — SMN protein deficiency from biallelic SMN1 mutation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene
- Mechanism source: FDA Zolgensma product page (<https://www.fda.gov/vaccines-blood-biologics/zolgensma>)
- Approval status: phase3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — SMN1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Therapeutic window is not wide; natural-history timing and irreversible damage should be reviewed
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: SRP-9001

Precedent disease: Duchenne muscular dystrophy

Vector: AAV9

Tissue target: muscle

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.71	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: muscle
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: myopathy
- Disease mechanism: loss of function — SMN protein deficiency from biallelic SMN1 mutation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene

- Mechanism source: FDA Zolgensma product page (<https://www.fda.gov/vaccines-blood-biologics/zolgensma>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — SMN1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Therapeutic window is not wide; natural-history timing and irreversible damage should be reviewed
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match

Dimension	Score	Max	What it measures
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (motor_neuron vs amino_acid_metabolism)
- Disease mechanism: loss of function — SMN protein deficiency from biallelic SMN1 mutation
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: FDA Zolgensma indication is SMA with biallelic SMN1 mutations; product supplies functional SMN transgene
- Mechanism source: FDA Zolgensma product page (<https://www.fda.gov/vaccines-blood-biologics/zolgensma>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — SMN1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multiple causal genes listed; score is gene-specific and should be repeated for each molecular subtype
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: muscle
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Therapeutic window is not wide; natural-history timing and irreversible damage should be reviewed
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone